

Exercise 3.3

Suppose $A \subset \mathbb{R}^n$ and let $(f_i)_{i=0}^\infty$ with $f_i : A \rightarrow \mathbb{R}^m$. Classify whether each statement is equivalent to (i) pointwise convergence $f_i \rightarrow f$, (ii) uniform convergence $f_i \rightarrow f$ uniformly, or (iii) neither.

- (a) Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $i \geq N$:

$$\sup_{x \in A} \|f_i(x) - f(x)\| < \varepsilon.$$

Classification: (ii) *uniform convergence*. This is exactly the definition via the sup norm.

- (b) For all $x \in A$, for all $i \in \mathbb{N}$ there exists $\varepsilon > 0$ such that

$$\|f_i(x) - f(x)\| < \varepsilon.$$

Classification: (iii) *neither*. The statement is always true (take, e.g., $\varepsilon = \|f_i(x) - f(x)\| + 1$), so it does not characterize convergence.

- (c) There exists $N \in \mathbb{N}$ such that for all $x \in A$, $i \geq N$, $\varepsilon > 0$:

$$\|f_i(x) - f(x)\| < \varepsilon.$$

Classification: (iii) *neither*. Since the left-hand side is nonnegative, the inequality for *all* $\varepsilon > 0$ forces $\|f_i(x) - f(x)\| = 0$; i.e. there is N with $f_i \equiv f$ on A for all $i \geq N$. This condition is much stronger than uniform convergence and is not equivalent to it in general.

- *d) $\forall \varepsilon > 0 \forall x \in A \exists N \in \mathbb{N} \forall i \geq N : \|f_i(x) - f(x)\| < \varepsilon$.

Classification: (i) *pointwise convergence*. The quantifier order matches the definition; N may depend on x and ε .

- *e) $\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall x \in A \forall i \geq N : \|f_i(x) - f(x)\| < \varepsilon$.

Classification: (ii) *uniform convergence*. Here N works simultaneously for all $x \in A$.

- *f) $\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall x \in A : (i \geq N \Leftrightarrow \|f_i(x) - f(x)\| < \varepsilon)$.

Classification: (iii) *neither*. This is far stronger than uniform convergence (it also demands that for $i < N$ the error is $\geq \varepsilon$ for every x), and it need not hold even when $f_i \rightarrow f$ uniformly.

Exercise 3.4

Suppose $A \subset \mathbb{R}^n$ and let $(f_i)_{i=0}^\infty$ with $f_i : A \rightarrow \mathbb{R}^m$ be a sequence of functions. Assume $f_i \rightarrow f$ *uniformly* on A . Show that if $B \subset A$, then $f_i \rightarrow f$ uniformly on B .

Solution. By uniform convergence on A , for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $i \geq N$ and all $x \in A$ we have $\|f_i(x) - f(x)\| < \varepsilon$. Since $B \subset A$, this inequality holds for all $x \in B$ as well, with the *same* N . Hence $f_i \rightarrow f$ uniformly on B .

Equivalently,

$$\sup_{x \in B} \|f_i(x) - f(x)\| \leq \sup_{x \in A} \|f_i(x) - f(x)\| \xrightarrow{i \rightarrow \infty} 0.$$